

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test – Bhaskara Contest

NMTC at JUNIOR LEVEL – IX & X Standards

Saturday, 1st September, 2018

J

Note:

1. Fill in the response sheet with your Name, Class and the institution through which you appear in the specified places.
 2. Diagrams are only visual aids; they are NOT drawn to scale.
 3. You are free to do rough work on separate sheets.
 4. Duration of the test: 2 pm to 4 pm – 2 hours.
-

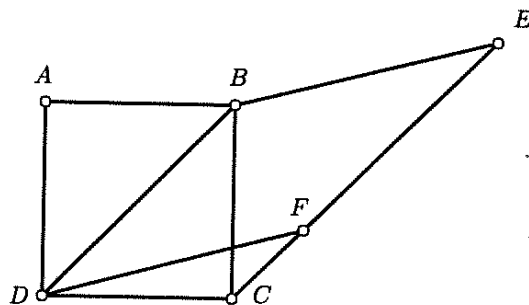
PART – A

Note

- Only one of the choices A, B, C, D is correct for each question. Shade the alphabet of your choice in the response sheet. If you have any doubt in the method of answering, seek the guidance of the supervisor.
- For each correct response you get 1 mark. For each incorrect response you lose $\frac{1}{2}$ mark.

1. The value of $\frac{3 + \sqrt{6}}{8\sqrt{3} - 2\sqrt{12} - \sqrt{32} + \sqrt{50} - \sqrt{27}}$ is
A. $\sqrt{2}$ B. $\sqrt{3}$ C. $\sqrt{6}$ D. $\sqrt{18}$
2. A train moving with a constant speed crosses a stationary pole in 4 seconds and a platform 75 m long in 9 seconds. The length of the train is (in meters)
A. 56 B. 58 C. 60 D. 62
3. One of the factors of $9x^2 - 4z^2 - 24xy + 16y^2 + 20y - 15x + 10$ is
A. $3x - 4y - 2z$ B. $3x + 4y - 2z$ C. $3x + 4y + 2z$ D. $3x - 4y + 2z$
4. The natural number which is subtracted from each of the four numbers 17, 31, 25, 47 to give four numbers in proportion is
A. 1 B. 2 C. 3 D. 4
5. The solution to the equation $5(3^x) + 3(5^x) = 510$ is
A. 2 B. 4 C. 5 D. No solution
6. If $(x + 1)^2 = x$, the value of $11x^3 + 8x^2 + 8x - 2$ is
A. 1 B. 2 C. 3 D. 4

7. There are two values of m for which the equation $4x^2 + mx + 8x + 9 = 0$ has only one solution for x . The sum of these two values of m is
 A. 1 B. 2 C. 3 D. 4
8. The number of zeros in the product of the first 100 natural numbers is
 A. 12 B. 15 C. 18 D. 24
9. The length of each side of a triangle is increased by 20% then the percentage increase of area is
 A. 60% B. 120% C. 80% D. 44%
10. The number of pairs of relatively prime positive integers (a, b) such that $\frac{a}{b} + \frac{15b}{4a}$ is an integer is
 A. 1 B. 2 C. 3 D. 4
11. The four digit number $8ab9$ is a perfect square. The value of $a^2 + b^2$ is
 A. 52 B. 62 C. 54 D. 68
12. a, b are positive real numbers such that $\frac{1}{a} + \frac{9}{b} = 1$. The smallest value of $a + b$ is
 A. 15 B. 16 C. 17 D. 18
13. a, b are real numbers. The least value of $a^2 + ab + b^2 - a - 2b$ is
 A. 1 B. 0 C. -1 D. 2
14. I is the incenter of a triangle ABC in which $\angle A = 80^\circ$. $\angle BIC =$
 A. 120° B. 110° C. 125° D. 130°
15. In the adjoining figure, $ABCD$ is a square and $DFEB$ is a rhombus. $\angle CDF =$



- A. 15° B. 18° C. 20° D. 30°

PART – B

Note

- Write the correct answer in the space provided in the response sheet.
 - For each correct response you get 1 mark. **For each incorrect response you lose $\frac{1}{4}$ mark.**
16. $ABCD$ is a square. E, F are points on BC, CD respectively and $\angle EAF = 45^\circ$. The value of $\frac{EF}{BE + DF}$ is _____
 17. The average of 5 consecutive natural numbers is 10. The sum of the second and fourth of these numbers is _____
 18. The number of natural numbers n for which $n^2 + 96$ is a perfect square is _____
 19. n is an integer and $\sqrt{\frac{3n-5}{n+1}}$ is also an integer. The sum of all such n is _____
 20. $\frac{a}{b}$ is a fraction where a, b have no common factors other than 1. b exceeds a by 3. If the numerator is increased by 7, the fraction is increased by unity. The value of $a + b$ is _____
 21. if $x = \sqrt[3]{2} + \frac{1}{\sqrt[3]{2}}$, then the value of $2x^3 - 6x$ is _____
 22. The angles of a heptagon are $160^\circ, 135^\circ, 185^\circ, 140^\circ, 125^\circ, x^\circ, x^\circ$. The value of x is _____
 23. ABC is a triangle and AD is its altitude. If $BD = 5DC$, then the value of $\frac{3(AB^2 - AC^2)}{BC^2}$ is _____
 24. A sphere is inscribed in a cube that has a surface area of 24 cm^2 . A second cube is then inscribed within the sphere. The surface area of the inner cube (in cm^2) is _____
 25. A positive integer n is a multiple of 7. If $\sqrt{n}$ lies between 15 and 16, the number of possible value(s) of n is _____
 26. The value of x which satisfies the equation $\frac{\sqrt{x+5} + \sqrt{x-16}}{\sqrt{x+5} - \sqrt{x-16}} = \frac{7}{3}$ is _____
 27. M men do a work in m days. If there had been N men more, the work would have been finished n days earlier, then the value of $\frac{m}{n} - \frac{M}{N}$ is _____
 28. The sum of the digits of a two number is 15. If the digits of the given number are reversed, the number is increased by the square of 3. The original number is _____
 29. When expanded the units place of $(3127)^{173}$ is _____
 30. If $a : (b + c) = 1 : 3$ and $c : (a + b) = 5 : 7$, then $b : (c + a)$ is _____